Group Assignment: Group of 3 students STC720-2026

The subsequent list of topics is not exhaustive and other related or similar topics may also be included.

Select one of the following topics (or titles):

Enhanced traditional monitoring schemes:

1. Shewhart-type control charts with runs-rules
2. EWMA control chart for monitoring skewed data
3. Synthetic EWMA control chart for financial time series
4. CUSUM control chart for monitoring health data
5. EWMA chart for pattern recognition
6. A pattern recognition CUSUM framework for monitoring complex data
7. Adaptive attribute EWMA control chart
8. Adaptive attribute CUSUM control chart
9. Multivariate Shewhart control chart using specialized runs-rules
10. Multivariate EWMA control chart for high-dimensional data
11. Multivariate CUSUM control chart for high contaminated data
12. Residual nonparametric control chart.
13. **Your own**.

Machine-Learning-based frameworks:

1. SVM-based EWMA framework for detecting abrupt shifts in the scale parameter
2. SVM-based CUSUM framework for detecting abrupt shifts in the scale and location parameters concurrently
3. A hybrid SVM-CNN based CUSUM control chart S control charts
4. Ensemble machine learning-based EWMA classifiers for anomaly detection
5. Machine learning-based Shewhart control charts for a simultaneous monitoring of the process mean and variance
6. Machine learning-based EWMA control charts for a simultaneous monitoring of the process mean and variance
7. Machine learning-based CUSUM control charts for a simultaneous monitoring of the process mean and variance
8. CNN-Support Vector Regression-based multivariate EWMA control chart for monitoring shifts in the mean vector and variance-covariance matrix concurrently
9. Support Vector Regression-based multivariate CUSUM control chart for monitoring shifts in the mean vector and variance-covariance matrix
10. Kolmogorov-Arnold Neural Network-based multivariate EWMA control chart
11. Hybrid SVM-Random Forest Neural Network Multivariate control charts for detecting subtle shifts in multivariate processes
12. Hybrid SVM-XGBoost based EWMA control chart for monitoring complex shifts in the process parameters
13. Shifts detection using machine learning control charts supplemented with runs-rules
14. Recurrent neural network control chart for monitoring time-series data
15. Machine learning control chart for monitoring classical time-series data
16. Machine learning control chart for monitoring financial time-series data
17. Machine leaning based control chart with application in financial industry
18. CUSUM control chart using artificial intelligence
19. EWMA control chart using random forest classifier for shifts interpretation
20. EWMA control chart using artificial intelligence
21. Adaptive reinforcement learning-based control charts
22. Machine learning-based pattern recognition monitoring scheme
23. Adaptive genetic algorithm EWMA control chart
24. Hybrid reinforcement learning-Genetic Algorithm based CUSUM control chart
25. Bayesian neural network control chart for monitoring shifts in dynamic processes
26. A comparison study of Machine learning and classical control charts with application in manufacturing industry
27. **Your own**.

Structure of the short article:

Title

Abstract (minimum 150 words, maximum 250 words)

Keywords: at least 3

1. Introduction (+/- 1 page)

   (Include the literature review on the topic: SPC &/or ML)

2. Mathematical background (max 3.5 pages)
   - 2.1 Existing chart(s) (+/- 1.5 page)
   - 2.2 Machine learning (max 2 pages)

3. Proposed SPC/ML-based control chart (max 3 pages)
4. Results & discussion (max 3 pages): (should include at least 1 table and at least 1 figure)
   - 4.1 Performance of the proposed method (max 2 pages)
   - 4.2 Performance comparison (1 page)

5. Illustrative example (application) (+/- 2 pages)
6. Conclusion and recommendations (+/- half a page)

Appendix (+/- 1.5 page)

References

***Weights*:**

Title, abstract and keywords= 10, Citations = 5, sufficiency = 5, Introduction = 15, Background = 20, Results, discussion and illustrative example (application) = 35, Conclusions and recommendations = 5, References = 5.

Total mark = 100

Note: The above weights can be changed at the discretion of the lecturer(s).

Instructions:

- Provide all the codes used to generate the results in the appendix.
- Each paper should have at least 1 table of the results. The title of the table should be given above the table.
- Each paper should have at least 1 figure (a figure can contain one or several graphs). Graphs must be unambiguous, have a descriptive title and axes must be clearly labelled. The title of the figure must be given below the figure.
- Tables and figures should be numbered.
- All equations should be numbered.
- You can use either WORD document or LATEX pdf document.
- Each group must submit (upload) the assignment on ClickUP before the due date 05 November. Assignments sent through email or in the lecturer's pigeon hole will be accepted.
- Since the assignment will be given over a long period of time, no medical certificates or affidavits will be accepted in case the assignment is not completed within the given time.
- The assignment should be submitted **BEFORE 05 NOVEMBER@23h59**.
- An assignment with a similarity rate of more than 20% will be accepted only if there is enough proof (or evidence) showing that the student did not copy the work from another source.
- For transparency, AI results, text, or guidance must be declared under ***a "generative AI use declaration"*** that should be added below the keywords, and the STC720 team will not accept a similarity rate of more than 20%. ***Failure to comply with this instruction will result in a lower mark***.

Plan

Week 1: Preliminary preparation	**Week 2: Summary of the contents**	**Week 3: Discussion and compilation**
Choose the topic and give a title	Summarise the contents of the articles.	Discuss the results and implement the method (i.e., control chart)
Read the articles of reference	Write the introduction and mathematical background.	Write the conclusion and recommendations
Search for relevant articles (at least 3 articles) and read them	Write the codes and do the computations.	Write the abstract and Proofread the entire article to fix the mistakes.
	Summarise the results in table(s). Present graph(s).	**TURNITIN AND SUBMIT BEFORE THURSDAY 05 NOVEMBER @23h59.**